AYMEN BOUCHALI

Full-Stack Engineer | Applied AI | Networks & Security

Blida, Algeria | +213 5 42 62 96 45 | aymenbouchali14@gmail.com | github.com/Bouchali-Aymen

SUMMARY

Full-stack engineer building production web platforms (Next.js, Firebase, Node.js, Laravel) with a strong background in applied machine learning - computer vision, reinforcement learning, and federated learning - and solid foundations in computer networks and cybersecurity.

EXPERIENCE

Full Stack Developer - Zed Academy
2024 - Present

Algiers, Algeria

- Develop learner experience, authentication, commerce, and admin features for a national EdTech platform on Next.js and Firebase.
- Contribute to the public marketing site (zedacademy.net) with Next.js and Tailwind CSS.

Graduate Researcher - M.Sc. Computer Systems & Networks
2024 - Jun 2026

Saad Dahleb University, Blida

- Master's thesis on privacy-preserving IoT malware detection using deep reinforcement learning (PPO, A2C, DQN) with hierarchical federated learning across client, fog, and cloud tiers.

Frontend Developer - Build Prompt
2024 - 2025

Remote, United Kingdom

- Shipped production UI for an AI document-intelligence SaaS using Next.js, Nuxt.js, and Tailwind CSS in Agile collaboration with product and backend teams.

Full Stack Developer - OwlStack
2024 - 2025

Boufarik, Blida

- Delivered client web applications end to end with Next.js, Nuxt.js, Laravel, Nest.js, and Node.js, focusing on performance, accessibility, and maintainable architecture.

Head of Cybersecurity - CSCC Club
Sep 2023 - Sep 2024

University Club, Blida

- Led workshops and mentoring; designed and organized a campus CTF and a national CTF with participants across Algeria (web exploitation, binary exploitation, reverse engineering, forensics).

Undergraduate Researcher - B.Sc. Computer Science
2023

Saad Dahleb University, Blida

- Built and evaluated a supervised learning pipeline for SQL injection detection, including data preparation, classical/neural models, and validation.

Freelance Developer
Before 2023

Remote

- Delivered client web projects covering UI implementation, backend integration, and feature delivery during undergraduate studies.

EDUCATION

M.Sc. Computer Systems and Networks
2024 - Jun 2026

Saad Dahleb University, Blida

B.Sc. Computer Science
2021 - 2023

Saad Dahleb University, Blida

SELECTED PROJECTS

DataPrep

Annotation platform for CV datasets (Next.js + FastAPI + YOLO): review detections, draw/correct bounding boxes, and export labels for training.

Zed Academy Platform & Landing

Production EdTech product (web.zedacademy.net) and marketing site (zedacademy.net) - Next.js, Firebase, Tailwind CSS.

Connected Algeria / Eventers / BuildPrompt / Mazraati

National event platform, event operations tooling, AI SaaS frontend, and farm operations system across Next.js and full-stack stacks.

SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Nuxt.js, Tailwind CSS
Backend & Data: Node.js, Express, Nest.js, FastAPI, Laravel, PHP, Python, Firebase, MongoDB, MySQL, PostgreSQL
AI / ML: Computer Vision, YOLO, CNNs, Supervised & Self-Supervised Learning, Reinforcement Learning, Federated Learning
Networks & Security: Routing & Switching, VPN, DNS, Web Security, Reverse Engineering, Digital Forensics, CTF Design, Linux
Practices: Git/GitHub, REST APIs, Agile, UI/UX